Class 18: Hedge Funds I

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Today's Question

Hedge fund fees have been falling for 20 years. Why do they still exist?

Positioning

Where We've Been	Where We Are	Where We're Going
Private equity	Hedge fund strategies and structure	Are hedge funds worth it?

Learning Objectives

By the end of today's class, you should be able to:

1. Categorize major hedge fund strategies (L/S equity, global macro, event-driven, etc.)
2. Explain the incentive effects of hedge fund fee structures
3. Describe how hedge fund "beta" has evolved over time

Matt Levine: Leaving the Enchanted Forest Is Scary

Last class we discussed PE/VC — the “Enchanted Forest” where companies can raise unlimited money from VCs who tolerate losses as long as growth continues.

Key tension:

- Private markets: patient capital, no quarterly earnings pressure
- Public markets: short-termism, buybacks, profitability expectations
- IPO = commitment to eventually make money?

Connection to today: Hedge funds sit at the intersection — they use public market instruments but with private-market-like structures (lock-ups, limited disclosure, flexible mandates).

Hedge Funds vs. Mutual Funds: Transparency & Investors

Hedge Fund	Mutual Fund
Limited liability partnerships with only minimal disclosure of strategy and portfolio composition	Regulations require public disclosure of strategy and portfolio composition
Less than 100 investors unless “qualified”	Number of investors is not limited

Hedge Funds vs. Mutual Funds: Strategy

Hedge Fund	Mutual Fund
Very flexible — funds can act opportunistically and make a wide range of investments	Predictable, stable strategies stated in prospectus
Often use shorting, leverage, options	Limited use of shorting, leverage, options

Hedge Funds vs. Mutual Funds: Fees & Liquidity

Hedge Fund	Mutual Fund
Traditionally “2-and-20”: 2% management fee + 20% of profits. In practice, avg. fees have fallen to **1.38% and 17.4%** (Preqin, 2024)	Fees are usually a fixed percentage of assets (avg. 0.5% for equity funds)
Often have lock-up periods, require advance redemption notices, or may be gated	Can move more easily into and out of a mutual fund

Hedge Fund Strategies

- **Directional**
  - Bets on the direction of markets, sectors, or asset classes
  - Significant market exposure ($\beta \neq 0$)
- **Non-directional**
  - Exploit temporary misalignments in relative valuation across securities
  - Buy one type of security and sell another
  - Market neutral ($\beta \approx 0$)

What are the implications for the beta of each strategy?

Directional Trading

In contrast to non-directional, market-neutral strategies, directional funds make large sectoral bets.

Examples: Global Macro funds (e.g. Soros' Quantum Hedge Fund) make large macro bets on currencies, commodities, sectors.

Non-Directional Trading

Relative value identifies rich vs. cheap based on a model of what prices "should be":

- Sometimes, economic models (e.g. Royal Dutch/Shell — dual-listed shares of the *same company* that traded at persistent price differences)
- Often, statistical models

Statistical Arbitrage:

- Use data mining techniques to uncover systematic pricing patterns
- Trade on divergence from normal
- Can involve high frequency trading over short holding periods

Statistical Arbitrage Example: Pairs Trading

Find stocks which move together:

- For a given pair, the relative value (p_1/p_2) provides a buy/sell signal
- High p_1/p_2 implies buy stock 2 and short stock 1

David Shaw (D.E. Shaw) argues: "*...human beings don't like to trade against human nature, which wants to buy stocks after they go up, not down.*"

Long-short strategy is **self-financing** and **market neutral**.

Pairs Trading Cookbook: Finding Pairs

Step A: Find stocks which move together.

1. For each stock, subtract off mean price, divide by standard deviation (normalized prices):

$$\tilde{p} = \frac{p - \overline{p}}{\sigma_p}$$

2. Calculate sum of squared deviations of normalized prices across all possible pairs of stocks ($N(N-1)/2$):

$$\sum (\tilde{p}_1 - \tilde{p}_2)^2$$

3. Each stock is paired with the stock for which the sum of squared errors is smallest.

What is the implication for paired stocks' β?

Pairs Trading Cookbook: Trading Rule

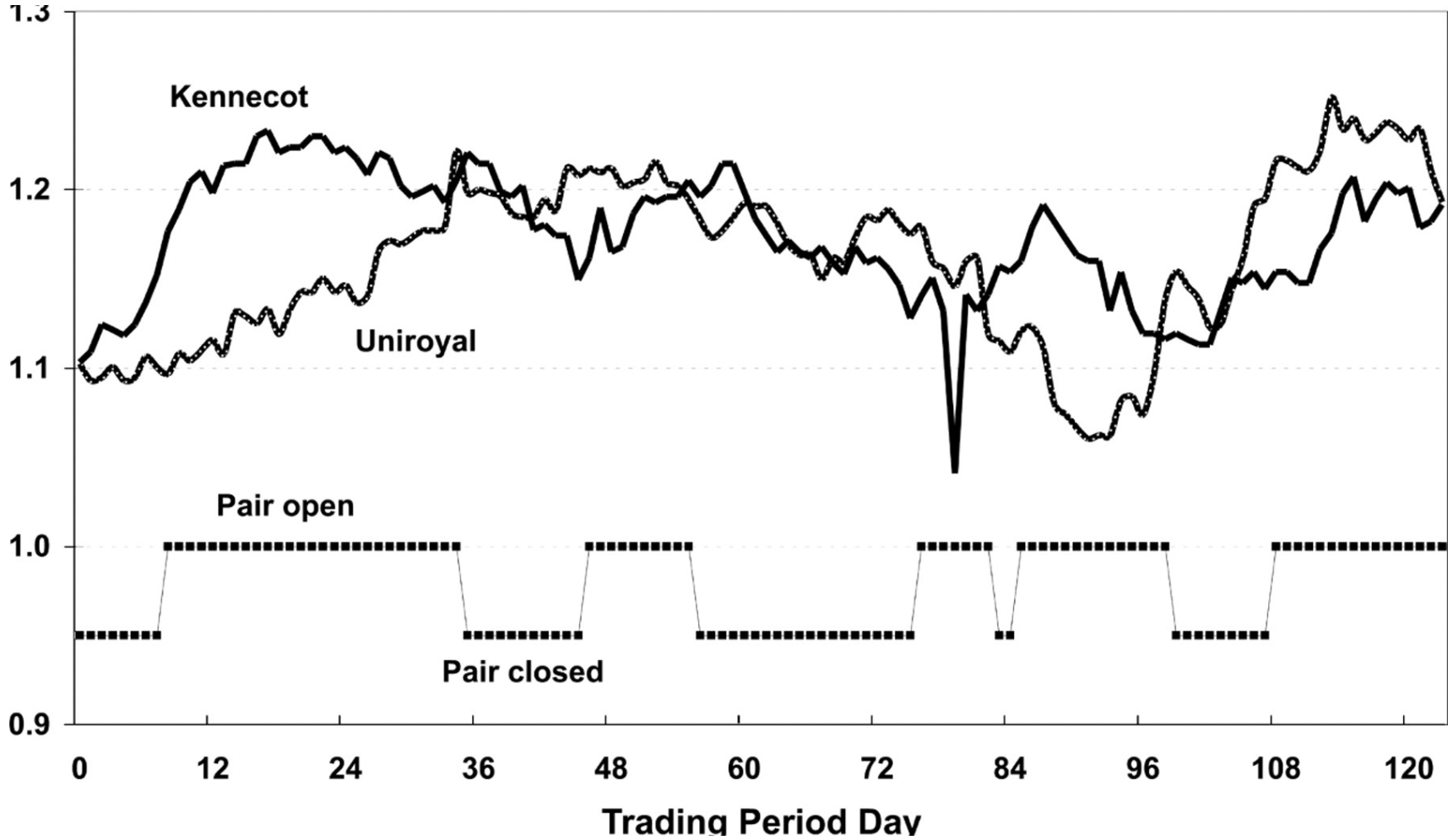
Limit focus to **"top pairs"**:

- Choose top 5% pairs based on minimum sum of squared deviations

Devise trading rule:

1. Define daily distance as $d = (\tilde{p}_1 - \tilde{p}_2)^2$ and calculate historical standard deviation of price differences
2. **"Open"** pair when $d > 2\sigma(d)$
3. **"Close"** pair when $d = 0$

Pairs Trading: Illustration



Pairs Trading Results

Excess returns of unrestricted pairs trading strategies

Pairs portfolio	Top 5	Top 20
A. Excess return distribution (no waiting)		
Average excess return (fully invested)	0.01308	0.01436
Standard error (Newey-West)	0.00148	0.00124
t-Statistic	8.84	11.56
Excess return distribution		
Median	0.01194	0.01235
Standard deviation	0.02280	0.01688
Skewness	0.62	1.39
Kurtosis	7.81	10.54
Minimum	–0.10573	–0.06629
Maximum	0.14716	0.13295
Observations with excess return < 0	26%	15%
Average excess return on committed capital	0.00784	0.00805

Pairs Trading: Factor Exposures

Does pairs trading expose the trade to risk factors?

Market neutrality suggests CAPM alpha, but what about other risks?

Factor model: Fama–French, Momentum, Reversal	
Intercept	0.00545 (3.81)
Market	−0.06661 (–1.03)
SMB	−0.04233 (–0.71)
HML	0.05740 (1.37)
Momentum	−0.02804 (–0.94)
Reversal	0.10192 (1.50)
R^2	0.05

Coefficients and t-statistics from a five-factor model.

Merger Arbitrage

Often a long-short strategy which buys the **target** and sells the **acquirer** in announced mergers.

Hedge fund holds **event risk**:

- Terms not reached
- Alternate bidders
- Regulatory approval not granted

Merger Arbitrage Example: Tellabs and Ciena

Tellabs and Ciena merger:

- Both companies involved in fiber-optic/telecom networks
- Merged company would be able to compete with larger firms such as Lucent by sharing customer bases, exploiting size

1-for-1 stock swap announced June 3, 1998:

- Tellabs: $64/share
- Ciena: $60/share
- Upon merger, prices must be equal

Merger Arbitrage Example: The LTCM Trade

Another infamous LTCM trade:

- Commit $30M own capital, borrow $30M @ 6% prime rate
- Buy 1M Ciena @ $60
- Sell 1M Tellabs @ $64
- Earn 4.5% short rebate (interest) on proceeds from short

What happens?

- Ciena loses two key customers, shares fall to $15, deal is abandoned
- Tellabs shares also decline before position is closed

Merger Arbitrage Example: Profit & Loss

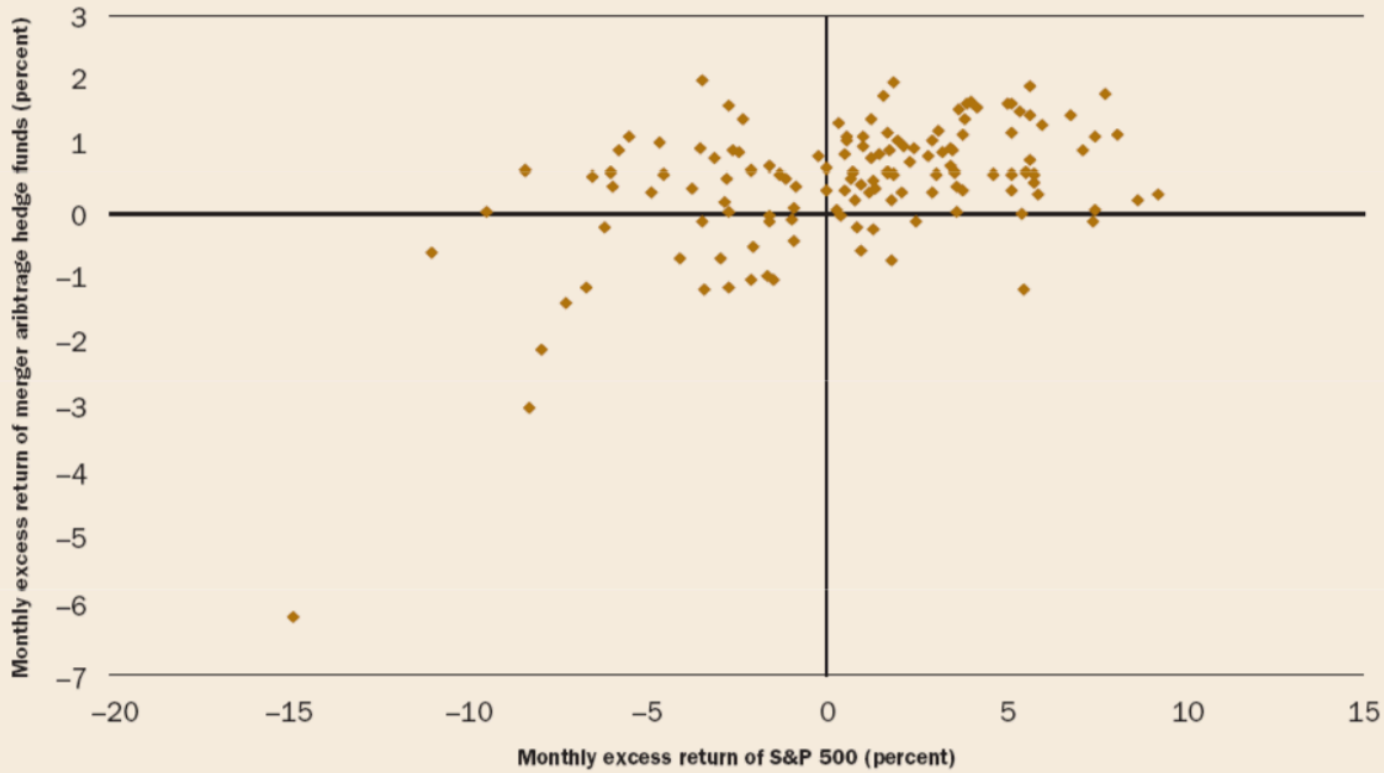
Profits and losses from trade:

Item	Amount
$30M @ 6% prime rate × (110/360)	–$550K
$64M short rebate @ 4.5% × (110/360)	$880K
Loss on Ciena: 1M × ($15 – $60)	–$45M
Gain on Tellabs: –1M × ($64 – $42)	$22M
TOTAL	**–$22.7M**

Total return on committed capital: –75.57%

Merger Arbitrage: Large Sample Evidence

Risk Factor for Merger Arbitrage Hedge Funds, 1994–2004



Source: HFR, Datastream

Merger Arbitrage: Profitability

Merger arb is profitable on average:

- Theory: hedge funds earn premium for insuring shareholders against completion risk

			Intercept		$R_{Mt} - R_f$		*SMB*		*HML*	
	N	R^2	a	p-value	b	p-value	s	p-value	h	p-value
Panel B: First offers										
Value-weighted										
Market model	192	0.15	0.86	(0.00)	0.32	(0.04)				
Fama-French three-factor	192	0.20	0.72	(0.02)	0.37	(0.02)	0.31	(0.04)	0.25	(0.07)
Equal-weighted										
Market model	192	0.12	0.80	(0.00)	0.22	(0.00)				
Fama-French three-factor	192	0.16	0.73	(0.00)	0.24	(0.00)	0.23	(0.01)	0.15	(0.11)

72–86 bps in monthly alpha!

How Has Hedge Fund 'Beta' Evolved?

Hedge funds were originally designed to **hedge** — Alfred Winslow Jones (1949) ran a market-neutral long/short equity fund.

Over time, the industry has shifted:

- Average hedge fund has equity $\beta \approx 0.37$ and bond $\beta \approx 0.14$ (Sullivan, 2021)
- Pre-GFC annualized α: **+3.4%/yr**; Post-GFC: **–0.8%/yr** (net of fees)
- Much of hedge fund "alpha" turned out to be repackaged factor exposure (Asness, Krail & Liew, 2001)

Implication: If you're paying 2-and-20 for beta you can get from an index fund at 5 bps, that's a bad deal. This helps explain why fees have been falling.

The Fee Squeeze in Numbers

Metric	Traditional	2024 Average
Management fee	2.00%	1.38%
Performance fee	20.0%	17.4%
Industry AUM	$200B (2000)	$4.88T

- Fees haven't been "2-and-20" in over a decade (Preqin, 2025)
- Yet AUM grew from $200B to $4.88T — total fee revenue still enormous
- Multi-strategy funds bucking the trend: mgmt fees **rising** due to complexity and pass-through costs

Sources: Preqin 2025 Global Report; Sullivan (2021), *J. Alternative Investments*

Next Class: April 9

Topics: Hedge Funds II

- Performance evaluation
- Dynamic factor loadings
- Non-linear risk exposures

Matt Levine Reading: “Hedge Funds are a Job Now”